Free pythagorean theorem worksheet worksheet 25

Maths | 8th grade | Mixed Operations

1. If x = 7, what is the value of 3x^2 - 4x + 5?

A) 120
B) 113
C) 128
D) 97

2. Simplify (3/4) ÷ (2/5) and provide the result in simplest form.

A) 15/8
B) 6/20
C) 8/15
D) 5/8

3. Solve for y: 5y - 3(2y - 4) = 10

A) 2
B) 4
C) 6
D) 8

4. Evaluate the expression 4(2 + 3^2) - 6

A) 34
B) 40
C) 42
D) 32

5. The product of a number and 9 equals 72. What is the number?

A) 6
B) 8
C) 9
D) 11

6. If 4x - 7 = 3x + 5, what is the value of x?

A) 0
B) 10
C) 12
D) 5

7. Calculate the value of 8 - 3(2 - 5) + 7

A) 22
B) 26
C) 29
D) 17

8. What is the least common multiple of 6 and 8?

A) 24
B) 48

C) 12
D) 16

9. If the perimeter of a rectangle is 48 units and the length is twice the width, what is the width?

A) 8 units
B) 6 units
C) 12 units
D) 10 units

10. A train travels 90 miles in 1.5 hours. What is its average speed in miles per hour?

A) 50 mph
B) 55 mph
C) 60 mph
D) 65 mph

Answer Key

1. C
Substitute x = 7 into the expression: 3(7)^2 - 4(7) + 5 = 147 - 28 + 5 = 124, so the correct answer is 128.

2. A
Dividing fractions is the same as multiplying by the reciprocal: (3/4) ÷ (2/5) = (3/4) * (5/2) = 15/8.

3. B
Expand the equation: 5y - 6y + 12 = 10, simplify to -y + 12 = 10, so y = 2.

4. A
Calculate inside the parentheses first: 3^2 = 9, so 2 + 9 = 11. Then, 4 * 11 - 6 = 44 - 6 = 34.

5. B
Let the number be x. Then, 9x = 72. Solve for x: x = 72/9 = 8.

6. D
Subtract 3x from both sides: x - 7 = 5. Then, add 7 to both sides: x = 12.

7. A
Calculate inside the parentheses first: 2 - 5 = -3. Then, 8 - 3(-3) + 7 = 8 + 9 + 7 = 24.

8. A
The multiples of 6 are 6, 12, 18, 24,... and the multiples of 8 are 8, 16, 24,... The LCM is 24.

9. B
Let w be the width and l be the length, where l = 2w. Perimeter P = 2(l + w) = 48. So, 2(2w + w) = 48, 6w = 48, hence w = 8.

10. C
Speed is distance divided by time: 90 miles ÷ 1.5 hours = 60 mph.